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Automatic Street Light Controller

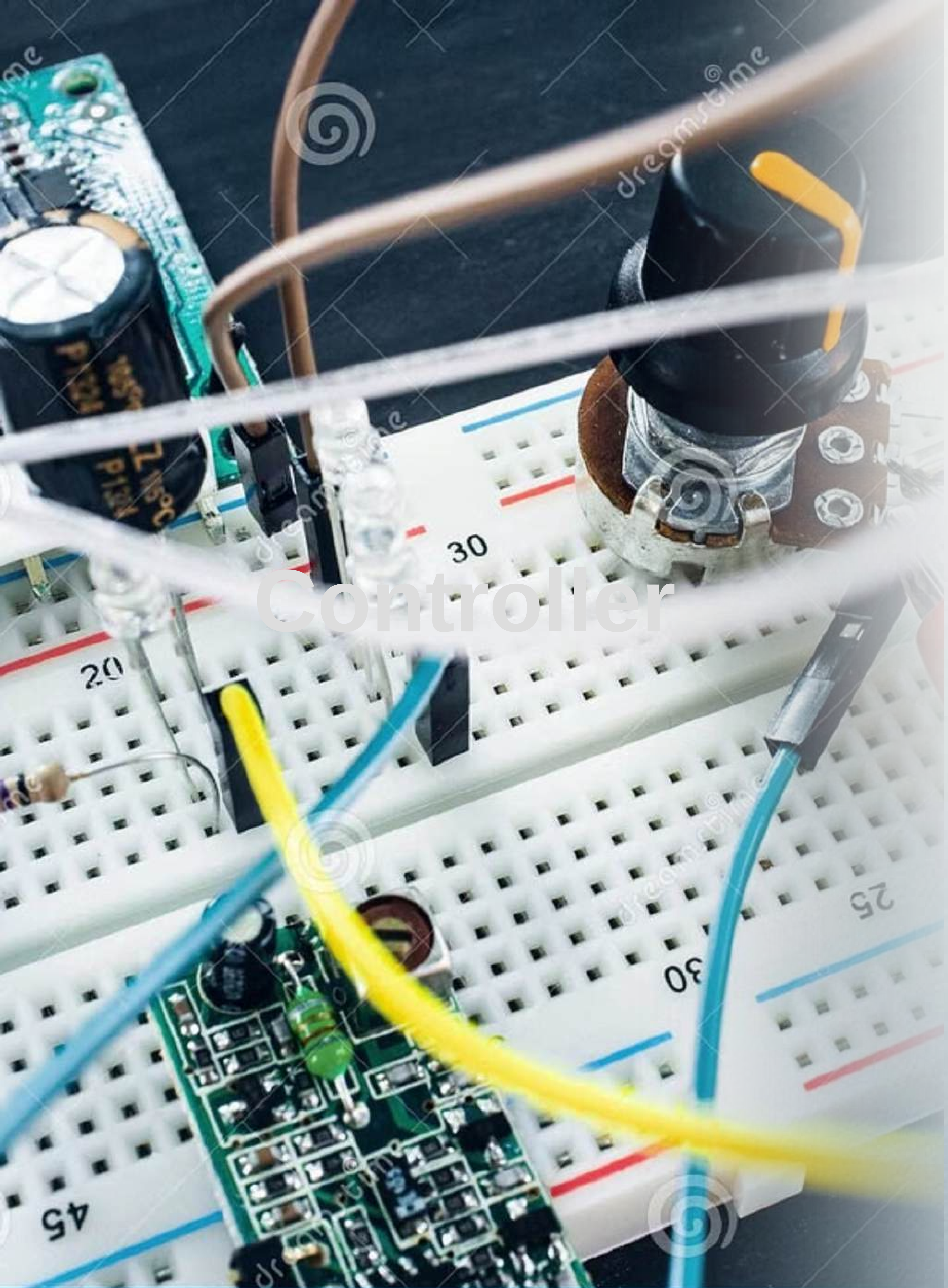
An automated system for efficient public and private lighting, eliminating manual switching, saving energy, and improving safety.

INPUT:

Uses an LDR(Light Dependent Resistor)Sensor to detect ambient light levels.

OUTPUT:

When it gets dark, a microcontroller turns on an LED or a street light.



Key Components & Architecture

Briefly identify the “players “in the circuit;

LDR Sensor

The sensor . Its resistance changes based on light intensity {high resistance in the dark, low resistance in the light}.

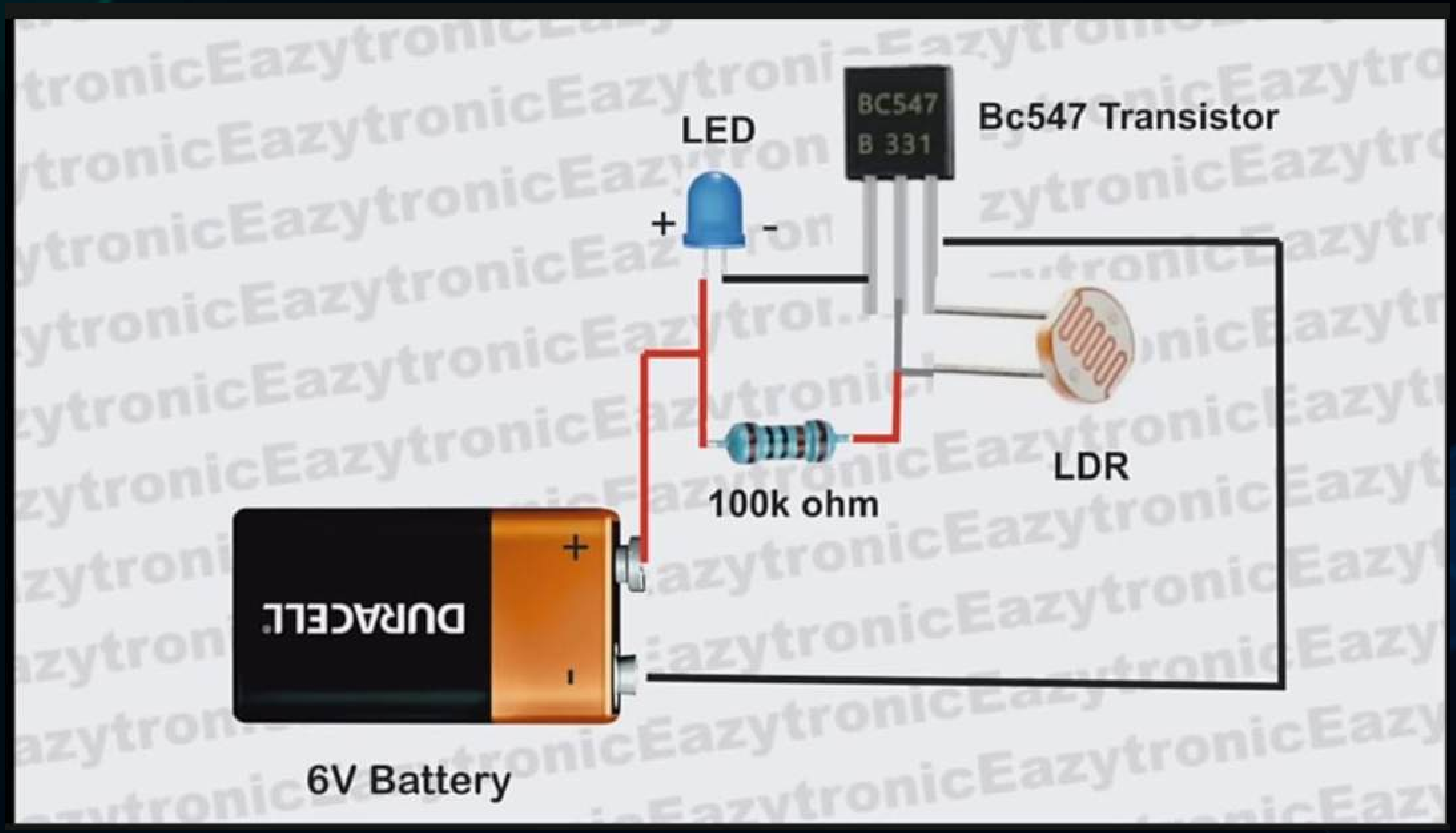
Bc547 Transistor

The “brain” or switch . It allows current to flow to the LED only when it receives a small signal at its “Base”(the middle pin “).

100 k ohm Resistor , LED

A 100K ohm resistor Controls the sensitivity of the sensor .

led controls high - power lamps; batteries or solar panels provide electrical energy.



BC547
B 331
Bc547 Transistor
LED
+
-
LDR
100k ohm
+
-
6V Battery

Core Concept & Functionality

SCENARIO A : IN BRIGHT LIGHT

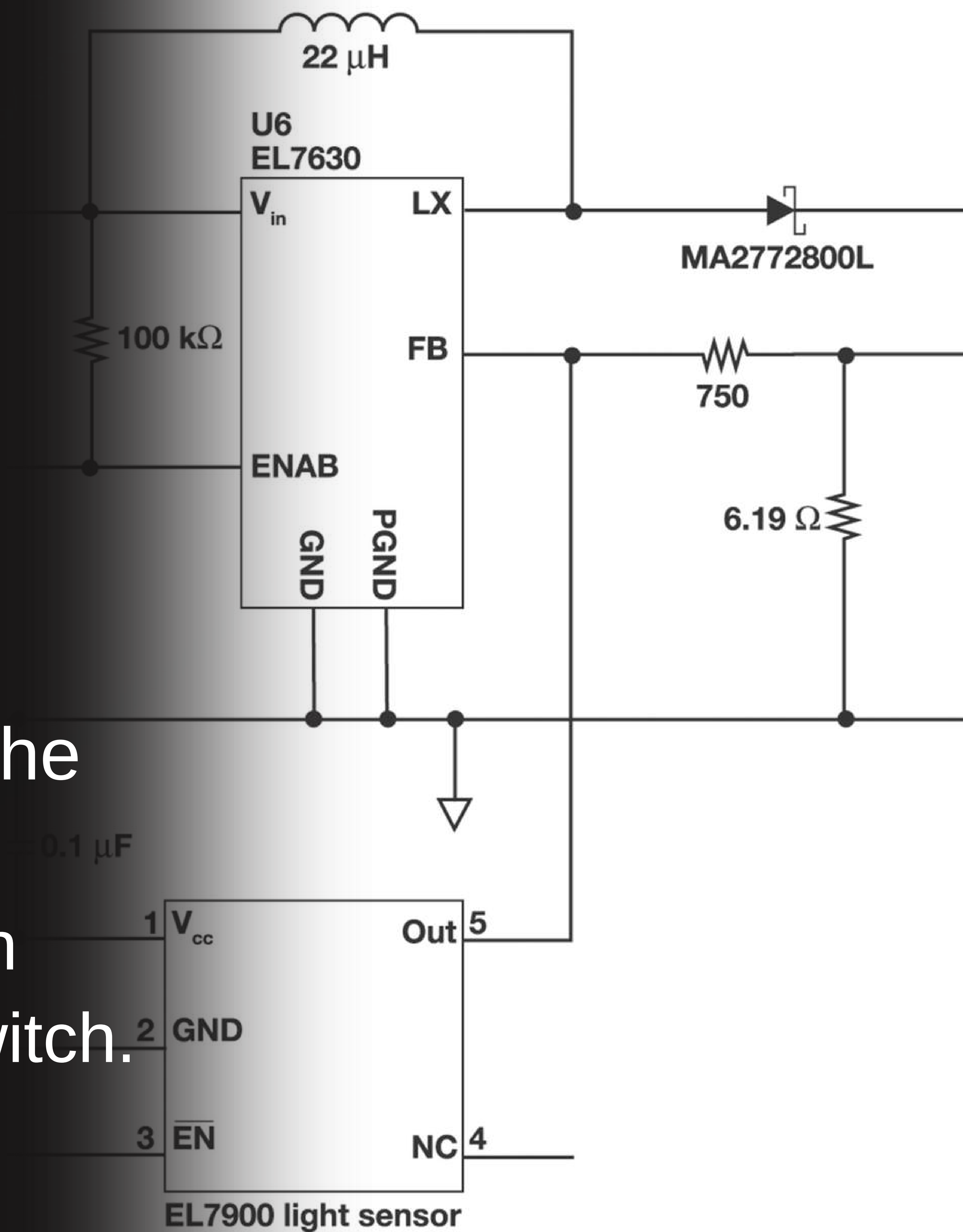
When light hits the LDR, its resistance drops significantly .

Ambient Light Based

The ‘drains’ the voltage away from the transistor’s base and send it straight to the negative terminal.

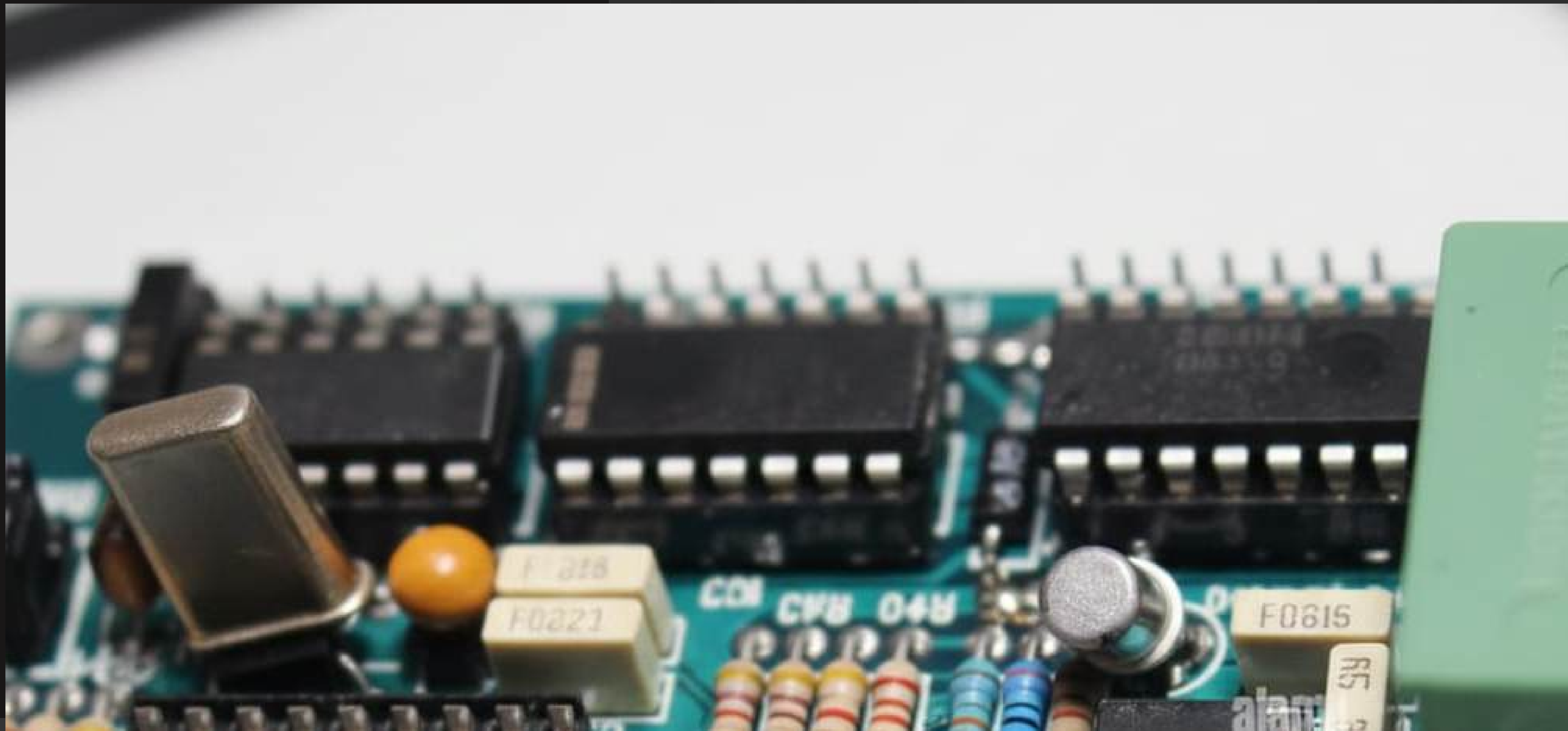
Because the transistor doesn't get enough voltage to ‘turn on ‘ , it says as an open switch.

RESULT: The LED stays ,OFF.”



SCENARIO B: IN DARKNESS

“When it gets dark , the LDR’s resistance becomes very high. Now, the electricity is forced through the 1ooK resistor into the base of the BC547 transistor . This triggers the transistor to ‘close the circuit ,’ allowing power to flow the battery through the LED .
The LED turns ON .”



Control Logic: Nightfall Sequence

01

Light Detection

Ambient light decreases, LDR resistance increases significantly.

02

Signal Processing

Voltage at microcontroller's analog input drops below threshold.

03

Switching Decision

Microcontroller identifies "darkness," sends HIGH logic signal.

04

Activation

Signal activates transistor, turning ON LED array, illuminating light.

Implementation Guide

1

Circuit Design

LDR with 10kΩ resistor for voltage divider. Logic-level MOSFET for power switching.

2

Microcontroller Programming

Use analogRead() for ADC. Implement hysteresis (±50 counts) and 1 Hz sampling rate.

3

LDR Placement

Clear, unobstructed sky view. Shield from direct LED light to prevent feedback loop.

4

Enclosure & Wiring

Weather-proof (IP65), adequate ventilation. Use appropriate wire gauge (14 AWG min).

5

Safety First

Install 10A fuse for overcurrent protection. Ensure proper earth connection.

Key Benefits of Automated Street Lighting

Implementing this system brings about significant advantages in energy conservation, operational efficiency, and overall public safety.

Energy Savings

Reduces electricity consumption by ensuring lights are only on when necessary.

Extended Lifespan

Prolongs the life of lighting fixtures by reducing unnecessary operating hours.

Enhanced Safety

Provides consistent and reliable lighting for pedestrians and motorists.

Reduced Maintenance

Automated operation minimizes the need for manual checks and adjustments.

Applications Beyond Street Lights

The fundamental principle of LDR-based automation extends beyond streets, finding utility in various other lighting scenarios.

- Garden and landscape lighting.
- Security and perimeter lighting for homes and businesses.
- Automated indoor lighting in large complexes.

Towards Smarter, Sustainable Illumination

The Automatic Street Light Controller is a simple yet powerful step towards building more efficient, eco-friendly, and responsive urban environments.

> "Innovation in lighting design is not just about brightness, but about intelligence and sustainability."

THANK YOU!

As we look to the future, these technologies will play a crucial role in shaping a better world for generations to come.